



Gowin Hardened MIPI C-PHY

User Guide

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Revision History

Date	Version	Description
02/27/2025	1.0E	Initial version published.
09/10/2025	1.0.1E	3.4 MIPI C-PHY IP TX Configuration updated and “mdclk” added for C-PHY TX PLL configuration.
01/05/2026	1.0.2E	● Updated Figure 2-2 MIPI C-PHY TX. ● Improved the configuration description in 3.1 MIPI C-PHY IP RX Configuration. ● Improved the configuration description in 3.4 MIPI C-PHY IP TX Configuration.
01/30/2026	1.0.3E	The functional descriptions, port definitions, and timing specifications for ALP mode added.

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1 About This Guide

1.1 Purpose

Gowin Hardened MIPI C-PHY User Guide mainly includes the description of the functions, features, ports, timing, and usage, etc. The guide helps users to quickly learn the feature and usage of Gowin Hardened MIPI C-PHY.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI Website. You can find the related documents at www.gowinsemi.com:

- [DS981, GW5AT series of FPGA Products Data Sheet](#)
- [DS1103, GW5A series of FPGA Products Data Sheet](#)
- [DS1118, GW5ART series of FPGA Products Data Sheet](#)
- [SUG100, Gowin Software User Guide](#)

1.3 Terminology and Abbreviations

The terminology and abbreviations used in this manual are as shown in Table 1-1.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
IP	Intellectual Property
LUT	Look-up Table
RAM	Random Access Memory
HS	High Speed
LP	Low Power

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

Gowin Hardened MIPI C-PHY complies with the MIPI C-PHY V1.2 standard. The document primarily includes the design features, functionality, port descriptions, configuration, and usage of MIPI C-PHY.

2.1 MIPI C-PHY Introduction

MIPI C-PHY is a high-efficiency high-speed serial interface protocol designed for short-distance communication. It utilizes a unique three-phase symbol encoding scheme to maximize data transmission efficiency. This encoding method not only ensures reliable data transmission but also maintains compatibility with various hardware designs.

MIPI C-PHY is widely used in mobile devices, automotive applications, and other embedded systems for high-speed data communication. Typical applications include camera interfaces, display panels, and sensor modules.

2.2 MIPI C-PHY Features

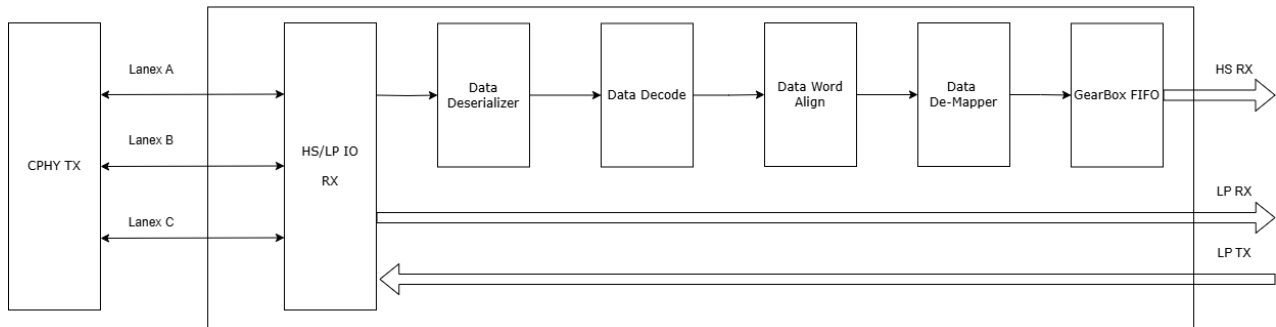
- **Compliance:** Compliant with the MIPI C-PHY V1.2 specification.
 - Core Structure: Includes a C-PHY hard core, supporting three channels, each consisting of a trio.
 - High-Speed Termination: Provides $100\ \Omega \pm 10\%$ differential impedance for high-speed receiver (RX) termination control.
- Supports RX/TX Combo-PHY, which can be configured as either RX or TX based on user requirements.
- **Low-Power Modes:** Supports MIPI LP-TX/RX signals, including RX-only and TX-only configurations.
- **High-Speed Signaling:**

- HS-RX/TX channel signals support a data rate of 800–2500 Msps.
- LP-TX/RX channel signals support data rates up to 10 Mbps.
- **Linear Equalization:** Provides a maximum delta peaking of > 8 dB @ 1.25 GHz.
- **Data Width:** Both TX and RX support 21-bit and 42-bit un-mapped data width, as well as 16-bit and 32-bit mapped data width.

2.3 MIPI C-PHY RX Function Description

MIPI C-PHY RX mainly includes HS/LP IO, serial-to-parallel conversion, data decoding, word alignment, data mapping, receive data FIFO, and other functional modules. The structural functions are shown in Figure 2-1.

Figure 2-1 MIPI C-PHY RX



HS/LP I/O

Supports dynamic ODT switching, LP TX/RX, and HS RX dynamic switching.

Data Deserializer

Deserializer is used to convert serial signals into parallel wire-state data.

Data Decoder

Decoder is used to convert wire-state data into symbol data.

Data Word Aligner

Word aligner is used to detect the sync word for data boundary detection.

Data De-mapper

De-mapper is used to convert symbol data into valid data.

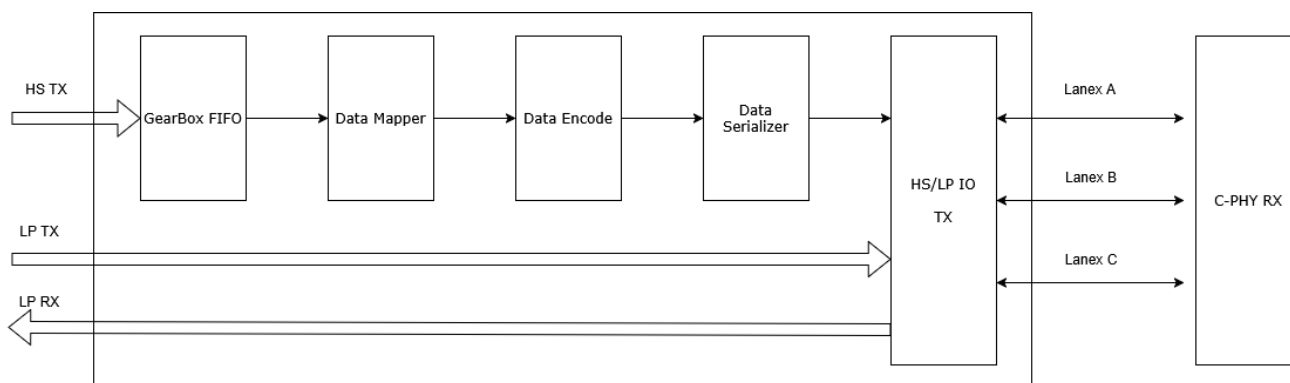
RX GearBox FIFO

GearBox FIFO is primarily used for RX asynchronous clock domain crossing and rate conversion.

2.4 MIPI C-PHY TX Function Description

MIPI C-PHY TX mainly includes HS/LP IO, parallel-to-serial conversion, data encoding, data mapping, and other functional modules. The structural functions are shown in Figure 2-2.

Figure 2-2 MIPI C-PHY TX



HS/LP I/O

- Supports data transmission with up to 3 data lanes, where each lane can be individually enabled or disabled.
- Supports dynamic switching between LP TX/RX and HS TX.

GearBox FIFO

GearBox FIFO is primarily used for TX asynchronous clock domain crossing and rate conversion.

Data Mapper

Mapper is primarily used to convert valid data into symbol data.

Data Encode

Encoder is used to convert symbol data into 3-wire-state.

Data Serializer

Serializer s used to convert parallel wire-state data into serial wire-state output.

Alternate Low-Power (ALP) mode

Alternate Low-Power Mode is an energy-saving operating mode in which unique code words are transmitted to replace the signaling levels on

high-voltage lines. This mode is designed to optimize link efficiency in low-power states while enabling rapid recovery to high-speed data transmission.

3 MIPI C-PHY IP Usage

Note!

No pin constraints are required when using the C-PHY hard IP.

3.1 MIPI C-PHY IP RX Configuration

Figure 3-1 MIPI C-PHY Configuration (RX)

The screenshot shows the 'MIPI_CPHY' IP Customization window. On the left is a schematic diagram of the IP block with various input and output pins. The right panel is titled 'General' and contains the following fields:

- Device: GW5AT-60
- Device Version: B
- Part Number: GW5AT-LV60UG225C2/I1
- Language: Verilog
- File Name: gowin_mipi_cphy_rx
- Module Name: Gowin_MIPI_CPHY_RX
- Create In: oreIP\Gowin_hardware_cphy\RefDesign\cphy_rx_2_5G_prbs9_32bit\src\gowin_mipi_cphy_rx ...

Below the 'General' tab are two tabs: 'RX' and 'TX'. The 'RX' tab is selected, showing the following configuration options:

- ☒ RX Enable
- ☒ Lane 0 Enable
 - Data Width: 42bit
 - Data A Reassign to: A
 - Data B Reassign to: B
 - Data C Reassign to: C
- ☒ Lane 1 Enable
 - Data Width: 42bit
 - Data A Reassign to: A
 - Data B Reassign to: B
 - Data C Reassign to: C
- ☒ Lane 2 Enable
 - Data Width: 42bit
 - Data A Reassign to: A
 - Data B Reassign to: B
 - Data C Reassign to: C
- RX Output Clock Selection: None

At the bottom right of the window are 'OK' and 'Cancel' buttons.

1. General Configuration

The General configuration is used to configure the information about the generated IP design file.

- Device: Display information about the configured Device.
- Device Version: Displays the version information of the configured Device.
- Part Number: Display the configured Part Number.
- Language: Hardware description language used to generate the IP design files. Click the drop-down list to select the language, including Verilog and VHDL.
- Module Name: The module name of the generated IP design files. It can be re-edited in the text box on the right. Module name cannot be the same as the primitive name. If it is the same, an error will be reported.
- File Name: The file name of the generated IP design files. It can be re-edited in the text box on the right.
- Create In: The path in which the generated IP files will be stored. Edit the target path in the right box or select the target path using the Select button on the right side of the text box.

2. RX Configuration

- RX_ENABLE: Enable RX configuration;
- Lane 0 Enable: Disable or enable lane 0.
- Lane 0 Data Width: 21bit or 42bit;
- Lane 0 Data A Reassign to: A/B/C, Default A;
- Lane 0 Data B Reassign to: A/B/C, Default B;
- Lane 0 Data C Reassign to: A/B/C, Default C;
- Lane 1 Enable: Disable or enable lane 1.
- Lane 1 Data Width: 21bit or 42bit;
- Lane 1 Data A Reassign to: A/B/C, Default A;
- Lane 1 Data B Reassign to: A/B/C, Default B;
- Lane 1 Data C Reassign to: A/B/C, Default C;
- Lane 2 Enable: Disable or enable lane 2.
- Lane 2 Data Width: 21bit or 42bit;
- Lane 2 Data A Reassign to: A/B/C, Default A;
- Lane 2 Data B Reassign to: A/B/C, Default B;
- Lane 2 Data C Reassign to: A/B/C, Default C.

3. RX Output Clock Selection: None/Lane0/Lane1/Lane2

- RX clock selection, choosing not to output the RX clock, or selecting a Lane recovery clock as the RX clock output

3.2 MIPI C-PHY IP RX Port

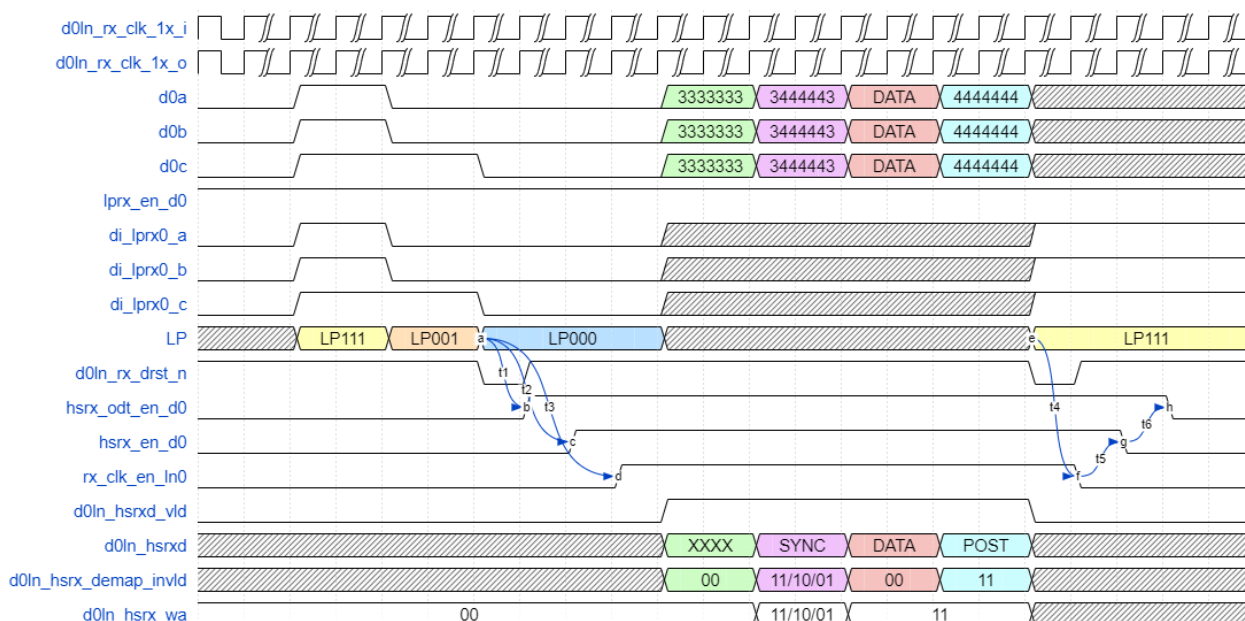
Table 3-1 C-PHY RX Port Description

Port	I/O	Description
d0ln_hsrxd	Output	Output data for lane 0. Refer to the bit map figure. [41:0]
d1ln_hsrxd	Output	Output data for lane 1. Refer to the bit map figure. [41:0]
d2ln_hsrxd	Output	Output data for lane 2. Refer to the bit map figure. [41:0]
d0ln_hsrxd_vld	Output	Data valid signal for lane 0. Preamble data is output as valid.
d1ln_hsrxd_vld	Output	Data valid signal for lane 1. Preamble data is output as valid.
d2ln_hsrxd_vld	Output	Data valid signal for lane 2. Preamble data is output as valid.
d0ln_hsrxd_demap_invid	Output	Not de-mapped data indication for lane 0. [1:0] -> [41:21, 20:0]
d1ln_hsrxd_demap_invid	Output	Not de-mapped data indication for lane 1. [1:0] -> [41:21, 20:0]
d2ln_hsrxd_demap_invid	Output	Not de-mapped data indication for lane 2. [1:0] -> [41:21, 20:0]
d0ln_hsrxd_fifo_rde_err	Output	FIFO read empty error indication for lane 0.
d1ln_hsrxd_fifo_rde_err	Output	FIFO read empty error indication for lane 1.
d2ln_hsrxd_fifo_rde_err	Output	FIFO read empty error indication for lane 2.
d0ln_hsrxd_fifo_wrf_err	Output	FIFO write full error indication for lane 0.
d1ln_hsrxd_fifo_wrf_err	Output	FIFO write full error indication for lane 1.
d2ln_hsrxd_fifo_wrf_err	Output	FIFO write full error indication for lane 2.
d0ln_hsrxd_wa	Output	Packet data alignment indication for lane 0.
d1ln_hsrxd_wa	Output	Packet data alignment indication for lane 1.
d2ln_hsrxd_wa	Output	Packet data alignment indication for lane 2.
d0ln_rx_clk_1x_o	Output	RX output 1x clock for lane 0.
d1ln_rx_clk_1x_o	Output	RX output 1x clock for lane 1.
d2ln_rx_clk_1x_o	Output	RX output 1x clock for lane 2.
Rx_clk_muxed	Output	RX output 1x clock. Selected by RX_OUTCLK_SEL
di_lprx0_a	Output	Lane 0 received LP-RX true data_a.
di_lprx0_b	Output	Lane 0 received LP-RX true data_b.
di_lprx0_c	Output	Lane 0 received LP-RX true data_c.
di_lprx1_a	Output	Lane 1 received LP-RX true data_a.
di_lprx1_b	Output	Lane 1 received LP-RX true data_b.

Port	I/O	Description
di_lprx1_c	Output	Lane 1 received LP-RX true data_c.
di_lprx2_a	Output	Lane 2 received LP-RX true data_a.
di_lprx2_b	Output	Lane 2 received LP-RX true data_b.
di_lprx2_c	Output	Lane 2 received LP-RX true data_c.
d0a	Inout	Data Lane 0 port a Input/Output.
d0b	Inout	Data Lane 0 port b Input/Output.
d0c	Inout	Data Lane 0 port c Input/Output.
d1a	Inout	Data Lane 1 port a Input/Output.
d1b	Inout	Data Lane 1 port b Input/Output.
d1c	Inout	Data Lane 1 port c Input/Output.
d2a	Inout	Data Lane 2 port a Input/Output.
d2b	Inout	Data Lane 2 port b Input/Output.
d2c	Inout	Data Lane 2 port c Input/Output.
d0ln_rx_clk_1x_i	Inputs	RX 1x clock from fabric for lane 0.
d1ln_rx_clk_1x_i	Input	RX 1x clock from fabric for lane 1.
d2ln_rx_clk_1x_i	Input	RX 1x clock from fabric for lane 2.
d0ln_rx_drst_n	Input	RX digital reset (active low) for lane 0.
d1ln_rx_drst_n	Input	RX digital reset (active low) for lane 1.
d2ln_rx_drst_n	Input	RX digital reset (active low) for lane 2.
lptx_enln0	Input	LP-TX enable for lane 0.
lptx_enln1	Input	LP-TX enable for lane 1.
lptx_enln2	Input	LP-TX enable for lane 2.
arst_rxln0	Input	Analog MIPI RX reset for lane 0. Active high.
arst_rxln1	Input	Analog MIPI RX reset for lane 1. Active high.
arst_rxln2	Input	Analog MIPI RX reset for lane 2. Active high.
do_lptx_a_ln0	Input	LP-TX line-a data for lane 0.
do_lptx_a_ln1	Input	LP-TX line-a data for lane 1.
do_lptx_a_ln2	Input	LP-TX line-a data for lane 2.
do_lptx_b_ln0	Input	LP-TX line-b data for lane 0.
do_lptx_b_ln1	Input	LP-TX line-b data for lane 1.
do_lptx_b_ln2	Input	LP-TX line-b data for lane 2.
alp_ed_out_ln 0	Output	Indicator signal: ● 1: ALP mode active ● 0: Default state
alp_ed_out_ln 1	Output	Indicator signal: ● 1: ALP mode active ● 0: Default state
alp_ed_out_ln 2	Output	Indicator signal: ● 1: ALP mode active ● 0: Default state

3.3 MIPI C-PHY RX Timing

Figure 3-2 MIPI C-PHY RX Timing



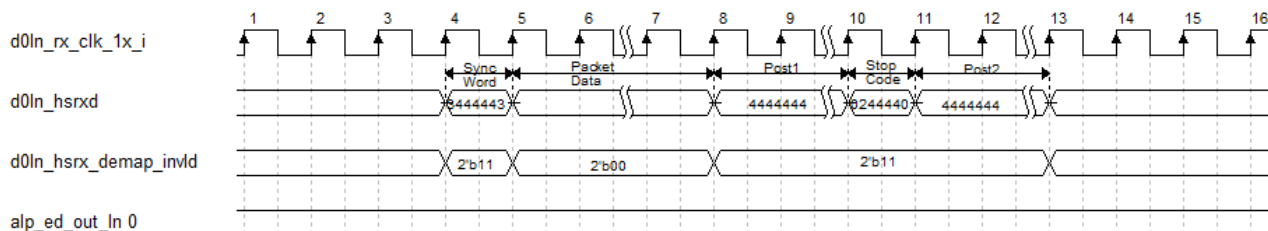
Timing requirements must be met:

1. $t1 \leq 38\text{ns}(t3 - \text{term} - \text{en})$
2. $t2 > t1$ and $t2 \leq 38\text{ns}(t3 - \text{term} - \text{en})$
3. $T3 > t2$ and $t3 > t3\text{-prepare}(38\text{ns} \sim 95\text{ns}) + 6\text{UI}$
4. $T4 > 0$, $t5 > 0$, $t6 > 0$

Note !

Ensure that $t4$ is long enough to allow complete reception of HS data. Do not pull it low immediately after detecting LP111.

Figure 3-3 ALP RX Timing



3.4 MIPI C-PHY IP TX Configuration

Figure 3-4 MIPI C-PHY Configuration (TX)

MIPI_CPHY

General

Device: GW5AT-60 Device Version: B
 Part Number: GW5AT-LV60UG225C2/I1 Language: Verilog
 File Name: gowin_mipi_cphy_tx Module Name: Gowin_MIPI_CPHY_TX
 Create In: D:\Gowin\HardCoreIP\Gowin_hardware_cphy\cphy_tx_2_5G_prbs9_32bit\src\gowin_mipi_cphy_tx

RX TX

☒ **TX Enable**

☒ **Lane 0 Enable**

Data A Reassign to: A
 Data B Reassign to: B
 Data C Reassign to: C

☒ **Lane 1 Enable**

Data A Reassign to: A
 Data B Reassign to: B
 Data C Reassign to: C

☒ **Lane 2 Enable**

Data A Reassign to: A
 Data B Reassign to: B
 Data C Reassign to: C

Data Width: 42bit
☐ Lane Divider Enable
 Lane Divider Ratio: 2

PLL Options

PLL CLKIN Frequency(MHz): 50.000 (19~800)
 mDRP Clock Frequency(MHz): 50 (1~100)
 MIPI CPHY Data Rate(Mbps): 2500 (800~2500)
 PLL VCO Frequency(MHz): 1250 (700~1400)
 CLKOUT0 Divider Factor: 1 Integer(1~128)
 0 Fractional(0~7)
 CLKOUT1 Divider Factor: 1 (1~128)
 CLKOUT2 Divider Factor: 1 (1~128)
 CLKOUT3 Divider Factor: 1 (1~128)
 CLKOUT4 Divider Factor: 4 (1~128)
 CLKOUT5 Divider Factor: 8 (1~128)
 CLKOUT6 Divider Factor: 8 (1~128)

Calculate

OK Cancel

1. General Configuration

The General configuration is used to configure the information about the generated IP design file.

- Device: Display information about the configured Device.
- Device Version: Displays the version information of the configured Device.
- Part Number: Display the configured Part Number.
- Language: Hardware description language used to generate the IP design files. Click the drop-down list to select the language, including Verilog and VHDL.

- **Module Name:** The module name of the generated IP design files. It can be re-edited in the text box on the right. Module name cannot be the same as the primitive name. If it is the same, an error will be reported.
- **File Name:** The file name of the generated IP design files. It can be re-edited in the text box on the right.
- **Create In:** The path in which the generated IP files will be stored. Edit the target path in the right box or select the target path using the Select button on the right side of the text box.

2. TX Configuration

- **TX_ENABLE:** enable TX configuration;
- **Lane 0 Enable:** Disable or enable lane 0.
- **Lane 0 Data A Reassign to:** A/B/C, Default A;
- **Lane 0 Data B Reassign to:** A/B/C, Default B;
- **Lane 0 Data C Reassign to:** A/B/C, Default C;
- **Lane 1 Enable:** Disable or enable lane 1.
- **Lane 1 Data A Reassign to:** A/B/C, Default A;
- **Lane 1 Data B Reassign to:** A/B/C, Default B;
- **Lane 1 Data C Reassign to:** A/B/C, Default C;
- **Lane 2 Enable:** Disable or enable lane 2.
- **Lane 2 Data A Reassign to:** A/B/C, Default A;
- **Lane 2 Data B Reassign to:** A/B/C, Default B;
- **Lane 2 Data C Reassign to:** A/B/C, Default C;
- **Data Width:** 21bit or 42bit;
- **Lane Divider Enable:** Enable or disable the Lane Divider;
- **Lane Divider Ratio:** 2~30, with a minimum adjustment step is 2;

3. PLL Option

- **PLL CLKIN Frequency(MHz):** Input range 19~800;
- **MIPI CPHY Data Rate (Mps):** Input range 800~2500;
- **PLL VCO Frequency(MHz):** Automatically calculated.
- **CLKOUT0 Divider Factor Integer:** Input range 1~128. For C-PHY simulation, this field must be set to 1 to ensure correct simulation results. If simulation is not required, any value may be used without affecting C-PHY operation.
- **CLKOUT0 Divider Factor Fractional:** Input range 0~7. For C-PHY

simulation, this field must be set to 0 to ensure correct simulation results. If simulation is not required, any value may be used without affecting C-PHY operation.

- CLKOUT1 Divider Factor: Input range 1~128. For C-PHY simulation, this field must be set to 1 to ensure correct simulation results. If simulation is not required, any value may be used without affecting C-PHY operation.
- CLKOUT2 Divider Factor: Input range 1~128. For C-PHY simulation, this field must be set to 1 to ensure correct simulation results. If simulation is not required, any value may be used without affecting C-PHY operation.
- CLKOUT3 Divider Factor: Input range 1~128. For C-PHY simulation, this field must be set to 1 to ensure correct simulation results. If simulation is not required, any value may be used without affecting C-PHY operation.
- CLKOUT4 Divider Factor: Input range 1~128. Any value within this range can be selected as required and will not affect C-PHY simulation or C-PHY operation.
- CLKOUT5 Divider Factor: Input range 1~128. Any value within this range can be selected as required and will not affect C-PHY simulation or C-PHY operation.
- CLKOUT6 Divider Factor: Input range 1~128. Any value within this range can be selected as required and will not affect C-PHY simulation or C-PHY operation.

The clock signal of the C-PHY hard IP is provided by PLL_VCO and is independent of the frequency settings of CLKOUT0–CLKOUT6. However, CLKOUT0–CLKOUT3 have an impact on C-PHY simulation.

For example, with a PLL CLKIN frequency of 50 MHz and a MIPI C-PHY data rate of 2500 Msps, the PLL VCO frequency is 1250 MHz, providing a 1250 MHz input clock for the C-PHY. To ensure correct C-PHY simulation, the divider factors of CLKOUT0–CLKOUT3 should be set to 1. CLKOUT4–CLKOUT6 can be configured to any frequency as required.

3.5 MIPI C-PHY IP TX Port

Table 3-2 C-PHY TX Port Description

Port	I/O	Description
clkin	Input	PLL clock input
mdclk	Input	PLL initial clock input
lock	Output	lock
clkout0	Output	PLL clkout0
clkout1	Output	PLL clkout1
clkout2	Output	PLL clkout2
clkout3	Output	PLL clkout3
clkout4	Output	PLL clkout4
clkout5	Output	PLL clkout5
clkout6	Output	PLL clkout6
spll0_ckp	Input	Serdes Clock0_P.
spll0_ckp	Input	Serdes Clock0_P.
spll1_ckp	Input	Serdes Clock1_P.
spll1_ckp	Input	Serdes Clock1_N.
Hstx_fifo_ae	Output	The TX FIFO write almost empty indication
Hstx_fifo_af	Output	TX FIFO read almost full indication
Hstx_fifo_rde_err	Output	3 lane FIFO read empty error indication(?)
Hstx_fifo_wrf_err	Output	3 lane FIFO write full error indication(?)
tx_clk_1x_o	Output	TX output 1x clock (max=178.57MHz)
di_lprx0_a	Output	Data lane0: Received LP RX data_a.
di_lprx0_b	Output	Data lane0: Received LP RX data_b.
di_lprx0_c	Output	Data lane0: Received LP RX data_c.
di_lprx1_a	Output	Data lane1: Received LP RX data_a.
di_lprx1_b	Output	Data lane1: Received LP RX data_b.
di_lprx1_c	Output	Data lane1: Received LP RX data_c.
di_lprx2_a	Output	Data lane2: Received LP RX data_a.
di_lprx2_b	Output	Data lane2: Received LP RX data_b.
di_lprx2_c	Output	Data lane2: Received LP RX data_c.
d0a	Inout	Data Lane 0 port a Input/output
d0b	Inout	Data Lane 0 port b Input/output
d0c	Inout	Data Lane 0 port c Input/output
d1a	Inout	Data Lane 1 port a Input/output
d1b	Inout	Data Lane 1 port b Input/output
d1c	Inout	Data Lane 1 port c Input/output
d2a	Inout	Data Lane 2 port a Input/output
d2b	Inout	Data Lane 2 port b Input/output
d2c	Inout	Data Lane 2 port c Input/output

Port	I/O	Description
clkin	Input	PLL input clock
d0ln_hstx_data	Input	HS data from fabric for lane 0, refer to the Bit map figure; [41:0]
d1ln_hstx_data	Input	HS data from fabric for lane 1, refer to the Bit map figure; [41:0]
d2ln_hstx_data	Input	HS data from fabric for lane 2, refer to the Bit map figure; [41:0]
d0ln_hstx_data_vld	Input	HS data valid from fabric for lane 0. Active high.
d1ln_hstx_data_vld	Input	HS data valid from fabric for lane 1. Active high.
d2ln_hstx_data_vld	Input	HS data valid from fabric for lane 2. Active high.
d0ln_hstx_map_dis	Input	Not mapped data indication signal for lane 0; [1:0] -> [41:21, 20:0]. Active high.
d1ln_hstx_map_dis	Input	Not mapped data indication signal for lane 0; [1:0] -> [41:21, 20:0]. Active high.
d2ln_hstx_map_dis	Input	Not mapped data indication signal for lane 0; [1:0] -> [41:21, 20:0]. Active high.
d0ln_tx_drst_n	Input	Data lane0 TX digital reset, Active Low.
d1ln_tx_drst_n	Input	Data lane1 TX digital reset, Active Low.
d2ln_tx_drst_n	Input	Data lane2 TX digital reset, Active Low.
hstx_enln0	Input	Data lane0: HS TX driver enable. Active high.
hstx_enln1	Input	Data lane1: HS TX driver enable. Active high.
hstx_enln2	Input	Data lane2: HS TX driver enable. Active high.
lptx_enln0	Input	Data Lane0: 1'b1 LPRX enabled. Active high.
lptx_enln1	Input	Data Lane1: 1'b1 LPRX enabled. Active high.
lptx_enln2	Input	Data Lane2: 1'b1 LPRX enabled. Active high.
Arstn_tx	Input	Reset of TX. Active Low.
do_lptx_a_ln0	Input	Data Lane0: LPTX line-a data
do_lptx_a_ln1	Input	Data Lane1: LPTX line-a data
do_lptx_a_ln2	Input	Data Lane2: LPTX line-a data
do_lptx_b_ln0	Input	Data Lane0: LPTX line-b data
do_lptx_b_ln1	Input	Data Lane1: LPTX line-b data
do_lptx_b_ln2	Input	Data Lane2: LPTX line-b data
do_lptx_c_ln0	Input	Data Lane0: LPTX line-c data
do_lptx_c_ln1	Input	Data Lane1: LPTX line-c data
do_lptx_c_ln2	Input	Data Lane2: LPTX line-c data
lprx_en_d0	Input	Data Lane0: LPRX enabled, Active High
lprx_en_d1	Input	Data Lane1: LPRX enabled, Active High
lprx_en_d2	Input	Data Lane2: LPRX enabled, Active High
alp0ln0	Input	Enable ALP: 1'b1 Default: 1'b0

Port	I/O	Description
alp0ln1	Input	Enable ALP: 1'b1 Default: 1'b0
alp0ln2	Input	Enable ALP: 1'b1 Default: 1'b0
alp0_impln 0	Input	Enable ALP: 2'b 10 or 2'b 11 Default: 2'b 00
alp0_impln 1	Input	Enable ALP: 2'b 10 or 2'b 11 Default: 2'b 00
alp0_impln 2	Input	Enable ALP: 2'b 10 or 2'b 11 Default: 2'b 00
alp_ed_enln 0	Input	Enable ALP: 1'b1 Default: 1'b0
alp_ed_enln 1	Input	Enable ALP: 1'b1 Default: 1'b0
alp_ed_enln 2	Input	Enable ALP: 1'b1 Default: 1'b0

3.6 MIPI C-PHY TX Timing

Figure 3-5 MIPI C-PHY TX Timing

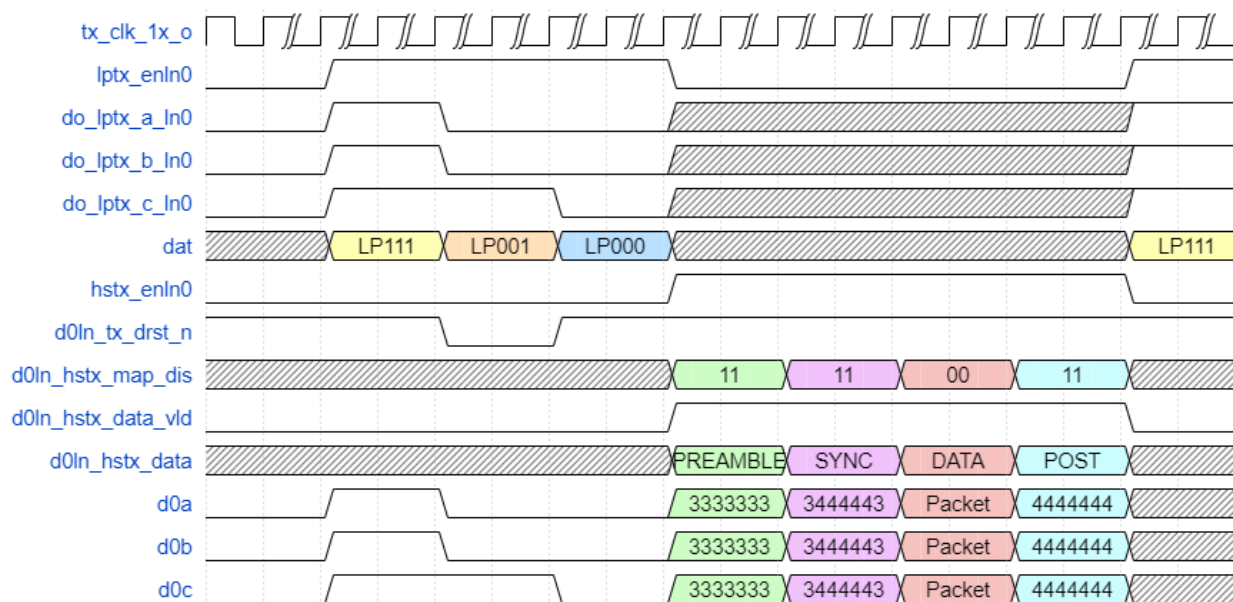
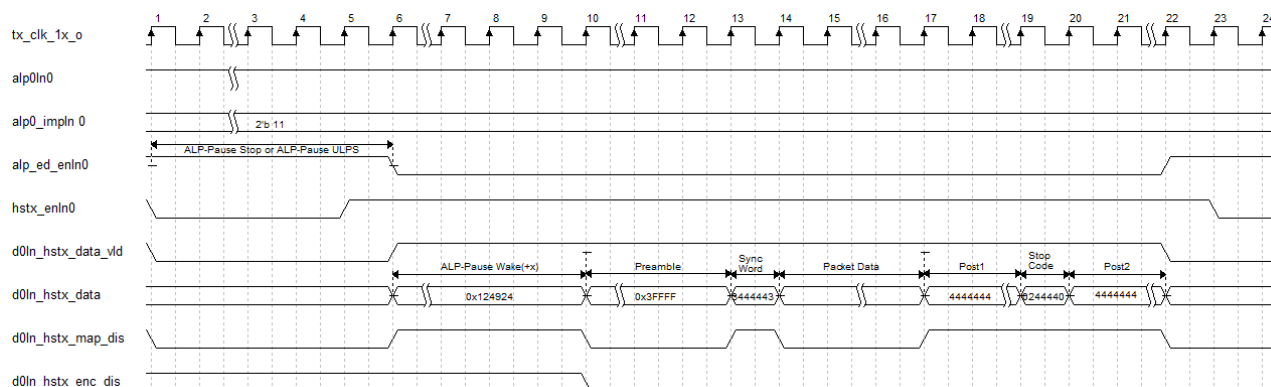


Figure 3-6 ALP TX Timing



ALP Operation Mode

During normal operation in ALP mode, the lane may be in the ALP-Pause state, ALP-Pause Wake state, or High-Speed data transmission mode. High-speed data transmission both begins and ends in the ALP-Pause state. The lane enters HS mode only during data burst or during ALP signals transmitting.

The procedure to enter HS mode for the data burst or ALP signal transmission is as follows:

1. Perform an ALP-Pause Wake operation, then transmit a Start Code consisting of either all-1 symbols or all-3 symbols.
2. The code words transmitted after the Start Code define the specific purpose of the burst. All code words used for ALP signaling have the same functions as their corresponding LP-mode states or escape-mode pulse sequences.
3. Upon detecting the Post2 code, the lane shall exit HS mode and return to the ALP-Pause state. If not in HS mode, the lane shall remain in the ALP-Pause state.

